LITERATURE REVIEW

Explainable AI for Short-Term Wind-Power Forecasting: Evaluating SHAP Explanation Stability Across Wind Regimes and Temporal Periods

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Abstract

Short-term wind-power forecasting has developed across statistical, machine-learning, deep-learning, hybrid, and data-driven approaches. This review synthesizes recent literature on 10-minute-scale forecasting, turbine SCADA data, Linear Regression (LR), XGBoost and Long Short-Term Memory (LSTM) models, wind-energy explainable artificial intelligence (XAI), SHAP attribution, explanation evaluation, and data provenance. The reviewed work already includes wind-specific SHAP applications, trustworthiness assessment, temporal explanation tools, and recent turbine-level forecasting comparisons. The research gap is therefore bounded: a reproducible assessment is needed of how SHAP attributions vary across predefined wind-speed regimes and chronological test windows, and how that variation relates to forecast accuracy when the models are evaluated on common observations and audited data. The proposed contribution is an empirical evaluation protocol, not a new forecasting algorithm, a causal explanation method, or a universal claim about trustworthy AI. No empirical findings are asserted in this review.

Keywords— wind-power forecasting, SCADA, XGBoost, LSTM, SHAP, explainable AI, explanation stability, chronological validation, data provenance.

1 Introduction and Scope

Wind generation is variable across time and operating conditions, making short-term forecasts relevant to scheduling, balancing, and the operational Integration of renewable generation. Recent reviews describe a diverse forecasting landscape that includes conventional statistical approaches, machine learning and deep-learning models, signal-processing pipelines, hybrid methods, and newer attention- or Transformer-based architectures [1]–[10]. They also show that reported accuracy depends on the forecast horizon, turbine or farm scale, available meteorological and operating inputs, preprocessing, feature design, and test protocol. Consequently, results from different sites and studies should not be read as a universal ranking of algorithms.

The present review is tailored to a 10-minute-ahead turbine active-power target and a common comparison of LR, XGBoost, and LSTM. The central methodological concern is to evaluate predictive accuracy and the reliability of model attributions together. Forecast results should be compared on the same held-out timestamps, with a common target definition and documented input availability. Recent wind-power evaluation research underscores that error metrics and reporting choices affect how studies can be interpreted and compared [11].

The review does not treat explainability as absent from wind research. Instead, it asks a narrower question: how can SHAP attribution variation be measured separately across operating regimes and across chronological windows within comparable regimes, and how does that variation align with forecast errors? SHAP outputs are interpreted as model attributions under specified assumptions; they are not treated as physical causes.

2 Wind-Power Forecasting and Selected Model Families

2.1 Recent forecasting approaches

Recent reviews organize short-term wind-power forecasting around data preparation, feature engineering, model choice, parameter tuning, and evaluation [1]–[10]. Individual studies illustrate this range: XGBoost has been optimized for wind-power prediction [12]; LSTM-based systems have been combined with preprocessing and optimization strategies [13]; and temporal-convolutional models have been paired with clustering or meteorological inputs [14]. The literature also includes attention- and Transformer-based designs, hybrid methods, and comparative studies using different forecast horizons. Section 2.4 summarizes recent comparator evidence. Because datasets, horizons, feature sets, and validation procedures vary, reported results do not support a universal ranking of algorithms.

2.2 Rationale for LR, XGBoost, and LSTM

The three-model design represents distinct forecasting roles. LR provides a transparent reference for testing whether more flexible predictors add value beyond a linear mapping. XGBoost supplies a boosted tree learner suited to structured SCADA features and nonlinear feature interactions; wind-specific work has examined its use and hyperparameter optimization [12]. LSTM provides a sequence-modeling comparator, and wind-power studies continue to adapt recurrent architectures to temporal inputs [13]. These choices make a compact comparison possible, but the review does not presume that one family will dominate for every turbine, horizon, or operating regime. Model performance must be established on the selected data and split rather than inferred from reports on other farms.

2.3 SCADA Inputs and Forecast-Time Availability

Wind-farm SCADA records combine active-power, meteorological, and turbine-operating channels, but usable feature sets depend on signal coverage and on whether each value is available at the forecast origin. Kelmarsh v4 provides multiyear 10-minute SCADA and event records [15], while the companion audit materials document reconstruction, signal onset, and quality flags for the recent files [16]. Penmanshiel provides a second 10-minute site with time-varying signal coverage [17]. SDWPF offers a distinct large-array setting [18], and recent cross-site work examines model-calibration portability [19]. For a 10-minute-ahead target, predictors should be derived strictly from readings available by the forecast issue time; post-issue observations must be excluded, and target alignment and any lag history should be stated explicitly. The selected wind-speed signal and regime variable should also be available at forecast time. Timestamp alignment, signal onset, event or quality flags, missingness, curtailment-related states, and reconstruction choices require an explicit audit because they can affect both forecast errors and SHAP distributions [15]–[17]. These resources motivate a documented feature-availability audit and careful distinction between within-site evaluation and external validation.

2.4 Recent Comparator Studies

Recent comparator studies include Transformer-based wind-power forecasting [20], an integrated boosting approach [21], and an LSTM method that explicitly reconstructs missing data [22]. Of particular relevance, a 2026 turbine-level comparison uses 10-minute data, lag- and rolling-based feature engineering, chronological splitting, and models including Ridge, Lasso, and XGBoost [23]. This is a close comparator for short-horizon model benchmarking, so the contribution should not be framed as the first turbine-level 10-minute comparison. Its reported scope does not, however, establish the proposed joint assessment of SHAP stability across wind regimes and chronological windows.

3 Data Provenance, Chronological Protocol, and Metrics

The proposed primary dataset is the Kelmarsh wind-farm v4 release, which provides 10-minute SCADA and event records for six turbines over multiple years [15]. For 2023–2024, the Park audit deposit supplies reconstructed files, a signal-onset manifest, and flags, and documents repeated cumulative blocks in the original exports [16]. The reconstruction and timestamp treatment should therefore be verified before modelling; a generic “keep last duplicate” rule should not be applied without checking the audit. The source release and any derived files should be tracked separately so that the data lineage remains reproducible.

Penmanshiel [17] can serve as a secondary site for external validation, while SDWPF [18] represents a different large-array context. Cross-site portability work [19] reinforces that transfer performance depends on site and calibration; results from a secondary dataset should be reported separately rather than pooled indiscriminately with Kelmarsh. Differences in turbine type, signal coverage, and operating distributions should be described before interpreting any apparent transfer.

The modelling protocol should use chronological training, validation, and final-test periods, with boundaries set only after confirming data integrity and channel coverage. Calculate metrics on common held-out timestamps: report MAE and RMSE in target units and R^2 as a supplementary measure; qualify comparisons across farms or differently scaled targets [11].

4 Explainable AI and SHAP in Wind Forecasting

Wind-energy XAI is already an established research area. Recent examples use SHAP to interpret data-driven ensemble predictions of turbine power [24] and to examine the contribution of meteorological factors to wind-farm power production [25]. [26]. directly examine trust in explainable wind-power forecasts and compare explanation approaches at global and instance levels [26]. A complementary line of work builds interpretability into a glass-box forecasting model rather than relying only on post-hoc attribution [27]. These studies establish that feature attribution and interpretability have already been applied to wind forecasting; a new study should not present XAI or SHAP itself as the research gap.

More recent work extends this evidence to recurrent and hybrid systems. An XAI-LSTM forecasting model uses SHAP to identify features associated with predictions [28], while a wind-energy review surveys model-agnostic and model-specific techniques and notes unresolved issues such as limited temporal explanation and benchmark-data limitations [29]. A system-level review of trustworthy renewable-energy forecasting treats explanation alongside robustness, data quality, and lifecycle considerations [30]. TimeSHAP has also been applied to a wind-turbine time-series model [31]. A 2026 multivariate wind-power study combines SHAP with a stacked bidirectional LSTM and reports repeated model-training rounds [32]; the distinction remains that reporting performance across initializations is not the same as measuring attribution-rank or attribution-magnitude stability across chronological windows and operating regimes. A physics-inspired hybrid forecasting study likewise uses SHAP for a residual model while assessing uncertainty [33].

In this study, SHAP should be used to summarize model behaviour relative to a stated background or reference sample. Global mean-absolute attributions, signed contributions, and individual explanations answer different questions and should not be conflated. Feature dependence and background selection can influence attribution allocation, so the feature set, reference data, model-specific explainer, and aggregation procedure should be fixed or explicitly varied in sensitivity checks [26], [29], [32].

5 Explanation Stability and Time-Series Evaluation

Recent time-series XAI work distinguishes explanations over time points, subsequences, or instances and highlights the difficulty of evaluating temporal explanations consistently [34]. A systematic review of XAI evaluation likewise treats explanation quality as multidimensional: stability, faithfulness, and other properties are related but not interchangeable, and no single score establishes that an explanation is reliable for every purpose [35]. Feature-attribution stability has been formalized and tested in broader settings [36], while later work shows that sampling and neighbour-selection choices can affect the repeatability of SHAP estimates [37]. These methods are not wind-specific, but they provide relevant methodological cautions for a SCADA application.

Time-series attribution methods also address the temporal structure of explanations. WindowSHAP groups consecutive time steps to reduce the explanation problem for long sequences [38], and evaluation work on multivariate time-series attribution compares methods using perturbation and task-based criteria [39]. Research on meaningful perturbations warns that unrealistic or out-of-distribution masking can produce misleading importance assessments [40]. Because these studies largely address time-series classification or other domains, their procedures should be adapted cautiously to forecasting rather than copied as if already validated for wind-power regression.

Operationally, keep four questions distinct: forecast accuracy, regime dependence, within-regime temporal stability, and the relationship between accuracy and stability. Compare LR, XGBoost, and LSTM on the same chronological held-out observations. For regime dependence, compare SHAP attribution profiles across prespecified low-, medium-, and high-wind-speed groups, with thresholds fixed from training data or a documented domain basis. For temporal stability, divide the held-out period into fixed chronological windows and compare each model's attribution ranks or vectors across windows within the same regime. Use pairwise Spearman rank agreement as the primary stability measure, with top-k overlap and attribution dispersion as secondary measures. For the accuracy–stability relationship, examine paired window-level error and stability summaries, including an exploratory Spearman association; do not assume a direction or infer causality. Before analysis, specify the forecast-time wind-speed signal, regime thresholds, window boundaries and minimum coverage, common feature mapping, model seeds, SHAP explainer, aggregation procedure, and a fixed training-data background/reference sample. Compute explanations on common test observations. A between-regime attribution difference represents condition dependence and is not, by itself, temporal instability.

A limited perturbation or grouped-ablation check can complement the stability metrics by testing whether highly attributed feature groups influence model output. The altered inputs should remain plausible for the SCADA data distribution; feature groups, replacement rules, and background samples should be specified in advance because arbitrary masking can create unrealistic inputs [37], [40]. These checks address aspects of model reliance or explanation faithfulness; they do not demonstrate that a sensor variable physically causes a power change.

6 Synthesis, Bounded Research Gap, and Novelty

The recent evidence establishes a substantial base for short-term wind-power forecasting, including boosted-tree, recurrent, convolutional, attention-based, and hybrid approaches [1]–[14], [20]–[23]. Wind specific explainability is also represented by SHAP studies, trustworthiness assessment, glass-box models, temporal explanation applications, and hybrid XAI work [24]–[33]. In addition, general time-series XAI research has proposed stability, faithfulness, and perturbation-based evaluation strategies [34]–[40]. The gap is therefore not a wholesale absence of wind-energy XAI, SHAP, or explanation-stability concepts.

Within the reviewed 2022–2026 sources, the narrower opportunity is to integrate four strands in one auditable experiment: comparative forecasting provides the LR, XGBoost, and LSTM model set; SCADA provenance and chronological validation protect the validity of the forecast test; wind-specific XAI supplies the application context for SHAP; and time-series explanation evaluation supplies candidate stability and faithfulness measures. The proposed experiment would use a common 10-minute-ahead turbine active-power target and common chronological test observations, quantify SHAP variation between predefined wind regimes and across chronological windows within comparable regimes, then examine how accuracy and explanation stability relate without presuming they co-vary. The Kelmarsh provenance audit makes explicit data reconstruction and signal-availability checks necessary [15], [16].

The defensible contribution is therefore a dataset-specific empirical assessment and reproducible evaluation protocol, not a new forecasting architecture, a causal explanation method, or a confirmed “first” claim. The distinction between condition dependence and temporal instability should be maintained throughout analysis and reporting. A final systematic search in scholarly databases is still needed before asserting that no closely matching regime-by-time SHAP study exists. Empirical answers to the research questions must be based on the study’s observed results; this review supplies no findings in advance.

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